

Homework1

Course: Quasilinear elliptic equations of second order

Problem 1

As we know in class: let $\alpha \in (0, 1)$ be a constant, Ω be a bounded domain in $\mathbb{R}^n$ satisfying an exterior sphere condition at $x_0 \in \partial\Omega$, and L be given by

$$Lu = a_{ij}\partial_{ij}u + b_i\partial_iu + cu \quad \text{in } \Omega,$$

for some $a_{ij}, b_i, c \in L^\infty(\Omega) \cap C(\Omega)$ satisfying

$$\lambda|\xi|^2 \leq a_{ij}(x)\xi_i\xi_j \leq \Lambda|\xi|^2 \quad \text{for any } x \in \Omega, \xi \in \mathbb{R}^n,$$

for some positive constants $\lambda \leq \Lambda$. Suppose that u is a $C(\bar{\Omega}) \cap C^2(\Omega)$ -solution of

$$Lu = f \quad \text{in } \Omega, \quad u = \phi \quad \text{on } \partial\Omega,$$

for some $f \in L^\infty(\Omega) \cap C(\Omega)$ and $\phi \in C(\partial\Omega)$ satisfying, for any $x \in \partial\Omega$,

$$|\phi(x) - \phi(x_0)| \leq \Phi_\alpha |x - x_0|^\alpha, \tag{1.1}$$

for some positive constant Φ_α . Then, for any $x \in \Omega$,

$$|u(x) - u(x_0)| \leq C \left(\sup_\Omega |u| + \Phi_\alpha + \sup_\Omega |f| \right) |x - x_0|^{\frac{\alpha}{1+\alpha}},$$

where C is a positive constant depending only on $n, \alpha, \lambda, \Lambda$, the L^∞ -norms of b_i and c , $\text{diam}(\Omega)$, and R in the exterior sphere condition.

Question: If we change the condition (1.1) to

$$|\phi(x) - \phi(x_0)| \leq \left| \frac{\Phi_\alpha}{\ln|x - x_0|} \right|^\alpha,$$

can we get the similar conclusion?